Waylon Jepsen

RUST SYSTEMS ENGINEER **waylonjepsen1@gmail.com**

Experience

Founding Performance Engineer, Pluto Labs Inc.
May 2024 - May 2025

- Proving TLS sessions with zero-knowledge cryptography.
- Built the fastest client side proving backend.
- Addressed performance needs by discovering circuit constraint optimizations in NIST cryptographic TLS candidates resulting in a 5x performance increase.
- Built elliptic curve pairing cryptographic primitives used widely.
- Optimized compile pipelines across WASM, Circom, and Rust, resulting in a 5x performance increase.
- contributed to the Circom compiler.

Founding Engineer, Primitive Bits
April 2022 - May 2024

- Built agent-based simulation modeling tools in Rust.
- Contributed to the core Rust Ethereum core infrastructure (reth, revm, and ethers-rs).
- Developed and maintained security and analysis tools for the EVM in Rust used by the larger community.

Network Engineer, Colorado State University
May 2021 - May 2022

- Built discrete event simulators to validate tropical algebraic TCP/IP Model.
- Developed Attack surface for VPNs via tropical algebra.

Systems Administrator, Colorado State University
July 2021 - April 2022

- Managed security, upgrades and backups for 600+ linux machines in the universities labs.
- Performed user database migrations to LDAP
- Worked on open cryptography problems concerning threshold cryptography and Shamir's Secret Sharing.

Education

Master of Science, Computer Science
Colorado State University, January 2020 - May 2022
Tropical Algebraic Modeling of TCP Congestion

Bachelor of Science, Mathematics & Philosophy
Colorado State University, August 2012 - May 2017
Elliptic Curve Cryptography

Research

1. W. Jepsen, *Replicating Portfolios: Constructing Permissionless Derivatives*, preprint, 2022. Available at arXiv: 2205.09890.

2. W. Jepsen *Cyclic Redundancy Checks and Error Detection*, preprint, 2022. Available arxiv: 2205.11344

Talks

Analysis of Constant Function Market Makers. DARPA Riser at Colorado State University, Sep 2022

Tropical Algebraic Analysis of TCP. Computer Science Seminar at Colorado State University, May 2022

Cyclic Redundancy Checks and Error Detection. Computer Science Seminar at Colorado State University, Aug 2021

The Lazarus Group: North Korea's Cyber War. Cyber Security Seminar at Colorado State University, May 2021

Epistemic Temporal Logic and Information flow Security. Cyber Security Seminar at Colorado State University, Jan 2021

The Mathematics of Elliptic Curve Cryptography. Cyber Security Seminar at Colorado State University, April 2019

Industry Involvement

Crypto Economics Security Conference (CESC'22)

Zero-Knowledge Proof Workshop (affiliated with CESC22)

ACM Conference on Economics and Computation (EC'22) *Optimism, Developer*

Advisory Board, Part time, November 2023

- Served on a technical board of advisors for optimism governance to ensure the technical quality of projects receiving funding from the optimism foundation.

Hacker Houses, Part time, September 2022 - 2024

- Raised over $250k in grants from Venture Capitalist
- Organized 6 hacker house series with over 2800 visitors in 3 continents
- Hosted 200 residential hackers who hacked 90 projects and raised more than $30M for their projects